① Aspect 4 (Highest Relevance)

Read, decode, and make sense of statements to create a model of the situation

The first requirement of this task is for students to understand a **real-world production context**:

- A company produces two types of electronic equipment (video and audio players)
- There is an average daily production quantity
- Each product type has a certain percentage of faulty items

Students must:

- Correctly interpret the written description
- Understand the meaning of quantities and percentages in the table
- Build a coherent mental model of "daily production → testing → faulty items"

This is a典型的 *(typical)* **formulation task involving constructing a situational model from text and tabular data**, making Aspect 4 the most central.

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② Aspect 6

Identify and describe the mathematical aspects of a real-world problem

From the context, students must identify key mathematical elements such as:

- Daily production quantities
- Fault rates (percentages)
- Numbers of faulty products (implicit variables)

Although no explicit formula is required, students must recognise that:

- The situation involves multiplicative reasoning (quantity × percentage)
- The real-world context has a clear underlying mathematical structure

③ Aspect 2

Identify the key variables in a model

Before judging whether the given statements are correct, students must clearly identify the model's key variables, including:

- Daily production of video players
- Daily production of audio players
- Fault rates for each type
- Daily numbers of faulty items

Correct identification of variables is essential for evaluating the statements.

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④ Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

Although the table is provided, students still need to:

- Choose an appropriate mathematical description to work with the data (e.g. using multiplication to find faulty quantities)
- Interpret percentages as multiplicative factors rather than standalone numbers

This reflects the ability to **select and use appropriate mathematical representations** within a given context.

⑤ Aspect 12

Identify assumptions, simplifications, and constraints in a model

The problem implicitly includes several modelling assumptions, such as:

- Production quantities and fault rates are averages
- The testing process is consistent for both product types
- Day-to-day variability is ignored

More advanced understanding recognises that:

- This is a simplified statistical model rather than an exact daily record

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⑥ Aspect 7

Simplify or decompose a situation

To evaluate the statements, students typically:

- Consider video and audio players separately
- Compute or reason about faulty quantities independently before comparing or combining results

This demonstrates the ability to **decompose a complex situation into manageable sub-problems**.

① Aspect 4（相关性最高）

Read, decode and make sense of statements … to create a model of the situation

📌 本题首先要求学生理解一个**真实生产情境**：

- 一家公司生产两种电子设备（video 和 audio players）
- 每天有固定的平均产量
- 每种产品都有一定比例的次品（faulty players）

学生必须：

- 正确解读文字说明
- 理解表格中“数量”和“百分比”的含义
- 在头脑中形成一个“每日生产—检测—次品”的整体模型

👉 这是一个典型的 **从文字与表格信息中构建情境模型** 的 formulate 任务，因此 Aspect 4 最为核心。

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② Aspect 6

Identify and describe the mathematical aspects of a real-world problem

📌 学生需要从情境中识别出关键的数学要素，例如：

- 每日生产数量
- 次品率（百分比）
- 次品数量（隐含变量）

👉 虽然题目没有明确要求建立公式，但学生必须意识到：

- 这是一个“数量 × 百分比”的问题
- 现实问题已经具备清晰的数学结构

③ Aspect 2

Identify the key variables in a model

📌 在判断三个陈述是否正确之前，学生必须清楚模型中有哪些关键变量：

- Video players 的日产量
- Audio players 的日产量
- 各自的故障率
- 每日故障产品的数量

👉 对变量的识别是后续判断陈述正确性的基础。

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④ Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

📌 题目已给出表格，但学生仍需要：

- 选择合适的"数学描述方式"来理解信息（如用乘法计算次品数量）
- 将"百分比"视为乘法因子，而非独立数值

👉 这体现了在给定情境中 **选择并使用合适数学表示** 的能力。

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⑤ Aspect 12

Identify constraints, assumptions, and simplifications in a model

📌 题目隐含若干模型假设，例如：

- 每天的生产数量和次品率是"平均值"
- Video 和 audio players 的检测标准一致
- 不考虑日与日之间的波动

👉 高层次理解会意识到：

- 这是一个简化的统计模型，而非精确的每日记录

⑥ Aspect 7

Simplify or decompose a situation

📌 为了判断陈述的正确性，学生通常会：

- 分别计算 video 和 audio 的次品数量
- 再进行比较或合并分析

👉 这体现了将复杂情境 **分解为可处理子问题** 的 formulate 能力。